A WEAKLY ETALE ALGEBRA OVER A FIELD IS REDUCED, INTEGRAL, A FILTERED COLIMIT OF FINITE SEPARABLE PRODUCTS, AND CLOSED UNDER SUBALGEBRAS, QUOTIENTS AND TENSOR PRODUCTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author numbers the parts as displayed, and in the proof the sentence 'Part (3) implies (2).' follows the discussion of part (3); that ordering is the author's and is reproduced. The author's phrasing 'any sub or quotient K-algebra' appears in the statement, reproduced unchanged. The six references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Characterisation of weakly etale algebras over a field as reduced algebras integral over the field whose finitely generated subalgebras are etale. invoked at proof lines 30864-30867 and 30873-30877. Is the criterion that turns the finitely generated description into weak etaleness for subalgebras.
- Etale K-algebras as finite products of finite separable extensions of K. invoked at proof lines 30865-30867. Supplies the shape of the finitely generated subalgebras and of the quotients whose colimits are taken.
- Permanence of weak etaleness under base change and composition. invoked at proof lines 30887-30888. Gives part (6) for tensor products.

Source reference. Original lemma 00U3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1097–1097. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30323: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30323–30323. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 30504: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30504–30504. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30303: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30303–30303. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30347: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30347–30347. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let K be a field. If B is weakly étale over K, then*

(1) *B is reduced,*
(2) *B is integral over K,*
(3) *any finitely generated K-subalgebra of B is a finite product of finite separable extensions of K,*
(4) *B is a field if and only if B does not have nontrivial idempotents and in this case it is a separable algebraic extension of K,*
(5) *any sub or quotient K-algebra of B is weakly étale over K,*
(6) *if B' is weakly étale over K, then $B \otimes_K B'$ is weakly étale over K.*

Proof. Part (1) follows from Lemma 30323 but of course it follows from part (3) as well. Part (3) follows from Lemma 30504 and the fact that étale K-algebras are finite products of finite separable extensions of K, see Algebra, Lemma 00U3. Part (3) implies (2). Part (4) follows from (3) as a product of fields is a field if and only if it has no nontrivial idempotents.

If $S \subset B$ is a subalgebra, then it is the filtered colimit of its finitely generated subalgebras which are all étale over K by the above and hence S is weakly étale over K by Lemma 30504. If $B \to Q$ is a quotient algebra, then Q is the filtered colimit of K-algebra quotients of finite products $\prod_{i \in I} L_i$ of finite separable extensions L_i/K. Such a quotient is of the form $\prod_{i \in J} L_i$ for some subset $J \subset I$ and hence the result holds for quotients by the same reasoning.

The statement on tensor products follows in a similar manner or by combining Lemmas 30303 and 30347. □